Cricbuzz LiveStats Project

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1. INTRODUCTION

The Cricbuzz LiveStats Dashboard is a data analytics project designed to analyze cricket match and player data using SQL and Python. The project provides an interactive platform where users can explore match insights, player statistics, and performance trends through a web-based dashboard.

The dashboard is developed using Streamlit and allows real-time interaction with structured cricket datasets. It integrates database operations, analytical queries, and visualization techniques to present meaningful insights.

🔗 Live Dashboard: Click Here

2. Objectives

The main objectives of this project are:

- To analyze cricket data using SQL queries
- To build an interactive dashboard for data visualization
- To perform CRUD operations on player data
- To generate insights on player and team performance
- To deploy the application for public access

3. Tools & Technologies

The following tools and technologies were used:

- Programming Language: Python
- Database: SQLite
- Libraries: Pandas, Streamlit
- IDE: VS Code
- Deployment: Streamlit CloudPandas
- VS Code

4. System Architecture

Data flows from SQLite database

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Processed using Python (Pandas)

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Visualized through Streamlit dashboard

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User interaction

The system follows a simple architecture:

- Data is stored in SQLite database
- Python connects to database using sqlite3
- SQL queries are executed for analysis
- Pandas processes the data

- Streamlit displays the data in dashboard

5. Architecture Diagram

Database (SQLite)

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Python (Pandas)

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Streamlit Dashboard

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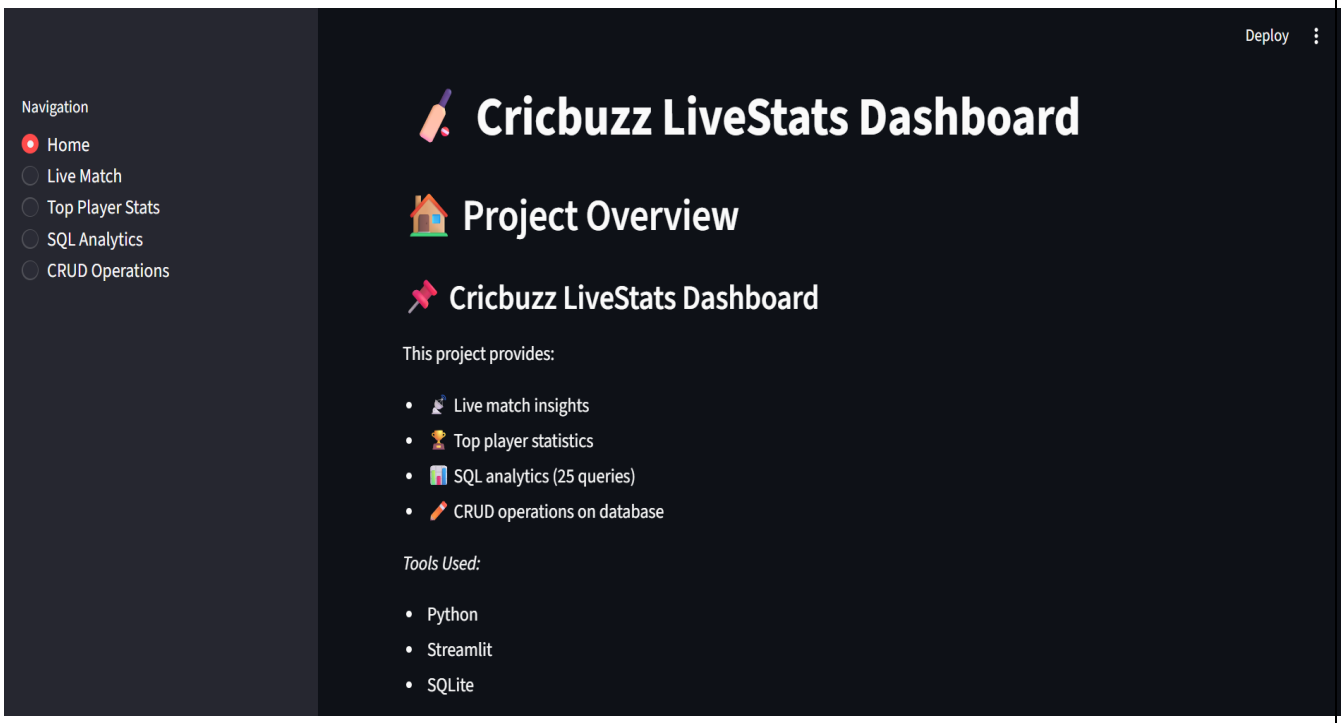
User Interaction

6. Dashboard Pages

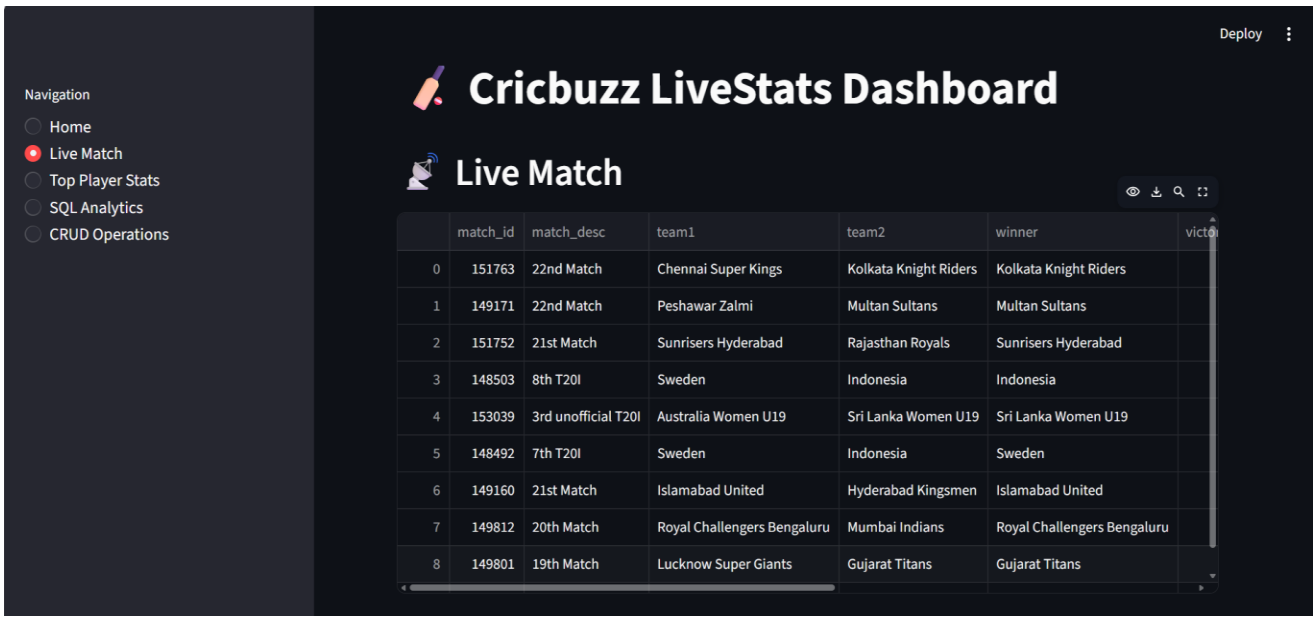
6.1 Home Page

Provides overview, navigation instructions, and project details.



6.2 Live Match Page

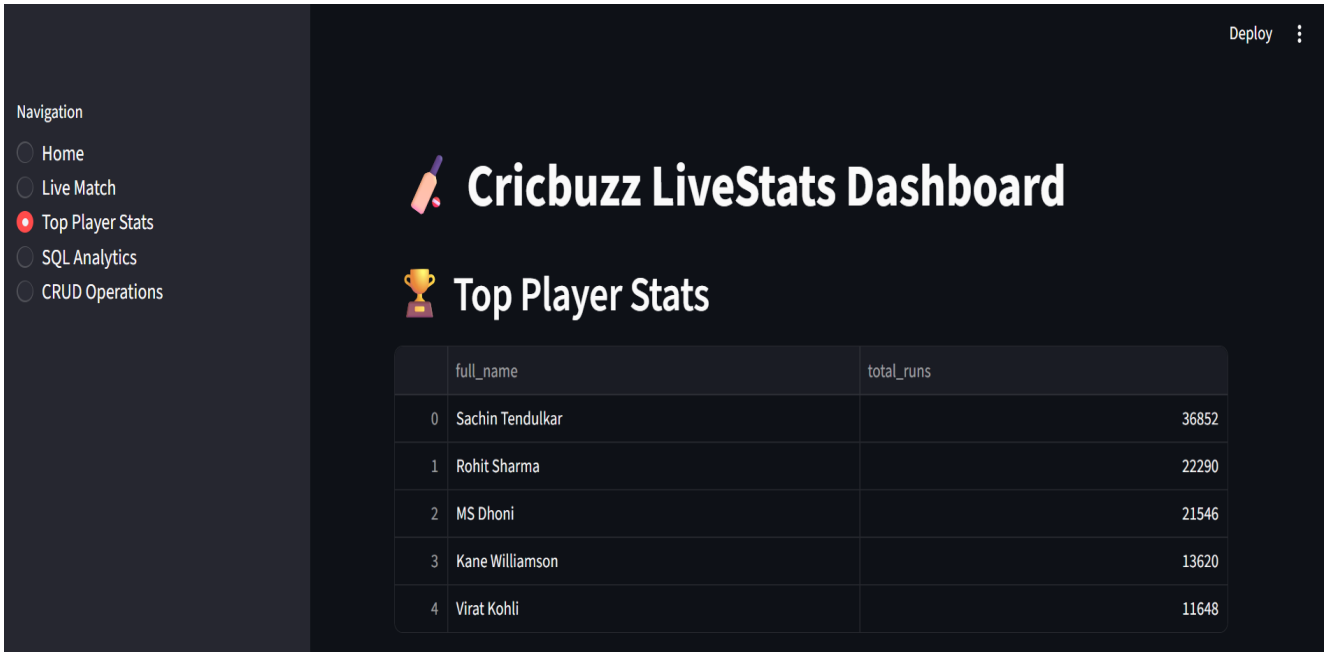
Displays match details including teams, venue, status, and results. Includes filters and summary metrics.



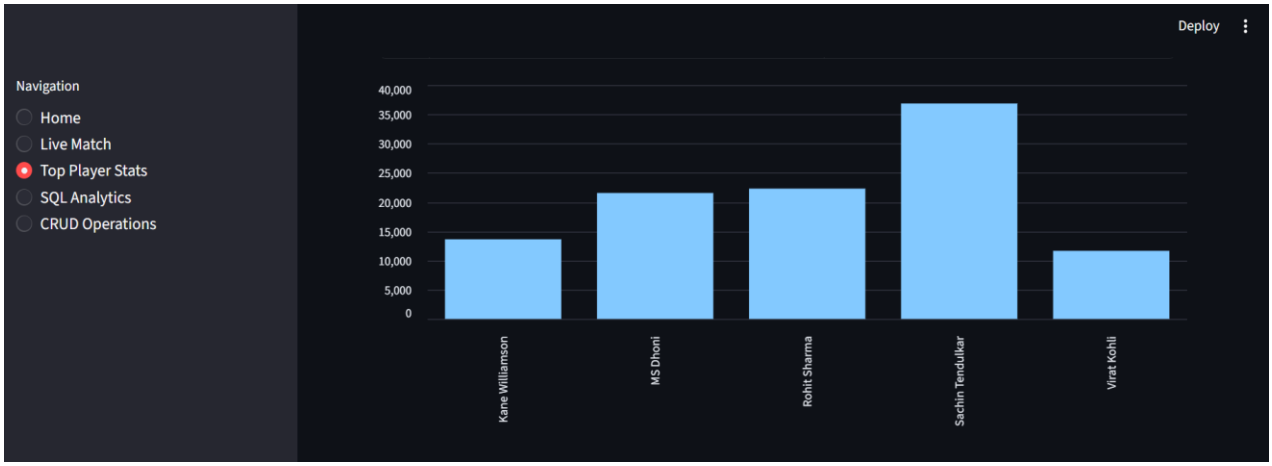
	match_id	match_desc	team1	team2	winner	victo
0	151763	22nd Match	Chennai Super Kings	Kolkata Knight Riders	Kolkata Knight Riders	
1	149171	22nd Match	Peshawar Zalmi	Multan Sultans	Multan Sultans	
2	151752	21st Match	Sunrisers Hyderabad	Rajasthan Royals	Sunrisers Hyderabad	
3	148503	8th T20I	Sweden	Indonesia	Indonesia	
4	153039	3rd unofficial T20I	Australia Women U19	Sri Lanka Women U19	Sri Lanka Women U19	
5	148492	7th T20I	Sweden	Indonesia	Sweden	
6	149160	21st Match	Islamabad United	Hyderabad Kingsmen	Islamabad United	
7	149812	20th Match	Royal Challengers Bengaluru	Mumbai Indians	Royal Challengers Bengaluru	
8	149801	19th Match	Lucknow Super Giants	Gujarat Titans	Gujarat Titans	

6.3 Top Player Stats Page

Shows top-performing players using charts and aggregated statistics.

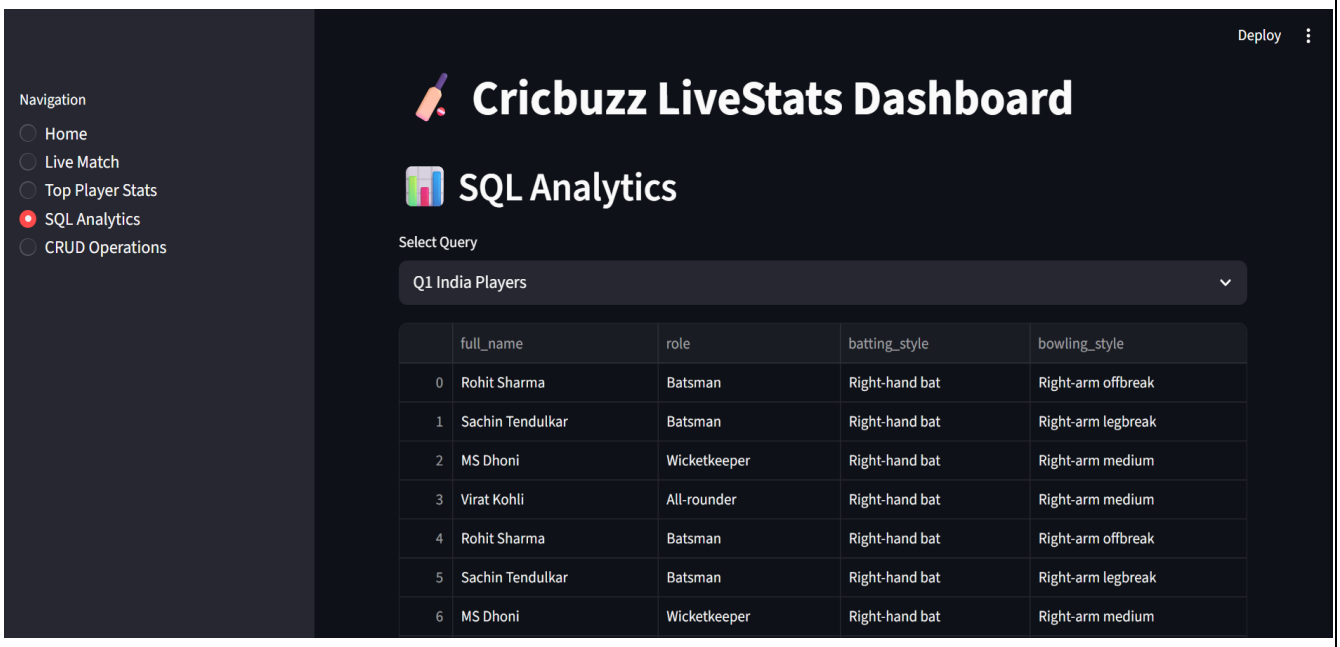


	full_name	total_runs
0	Sachin Tendulkar	36852
1	Rohit Sharma	22290
2	MS Dhoni	21546
3	Kane Williamson	13620
4	Virat Kohli	11648



6.4 SQL Analytics Page

Contains 25 advanced SQL queries providing insights into performance, trends, and match analytics.

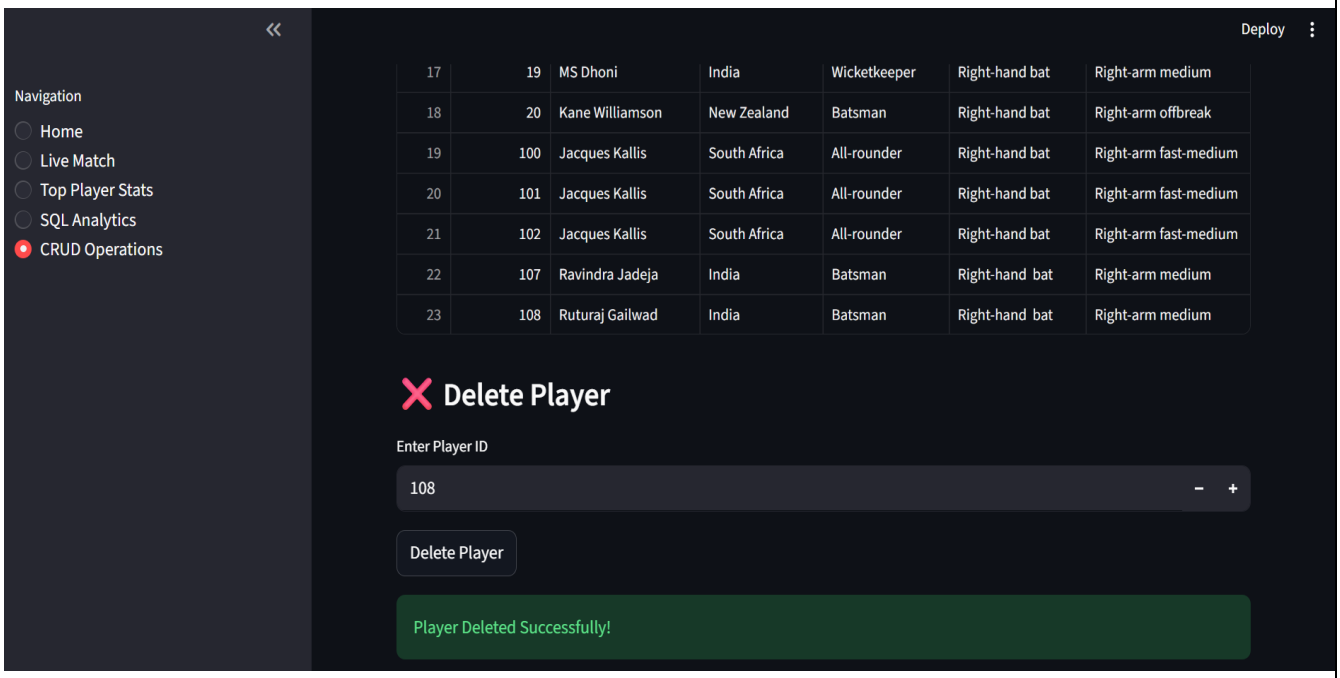


6.5 CRUD Operations Page

Implements Create, Read, Update, and Delete operations on player data.

ADD OPERATION

DELETE OPERATION



7. KEY SQL INSIGHTS

Some important insights derived:

- Top run scorers in ODI cricket
- Most economical bowlers
- Player consistency analysis
- Toss impact on match outcomes
- Team performance comparison
- Player ranking based on combined metrics

8. CHALLENGES AND SOLUTIONS

Challenge	Solution
Indentation errors in Python	Fixed proper code structure
Query integration in dashboard	Mapped queries using selectbox
CRUD not working	Corrected database queries
Deployment issues	Fixed file naming and requirements
Data inconsistency	Cleaned and validated dataset

9. LIVE DASHBOARD ACCESS

🔗 Live Dashboard: https://cricbuzz-livestats-btyfcjkk5zykby36wdanrp.streamlit.app/

The dashboard is deployed using Streamlit Cloud and provides real-time interaction with the dataset.

10. CONCLUSION

This project successfully demonstrates the integration of SQL, Python, and Streamlit to build a fully functional analytics dashboard. It provides meaningful insights into cricket data while also enabling real-time database operations.

Through this project, practical knowledge of data analysis, dashboard development, and deployment has been achieved.